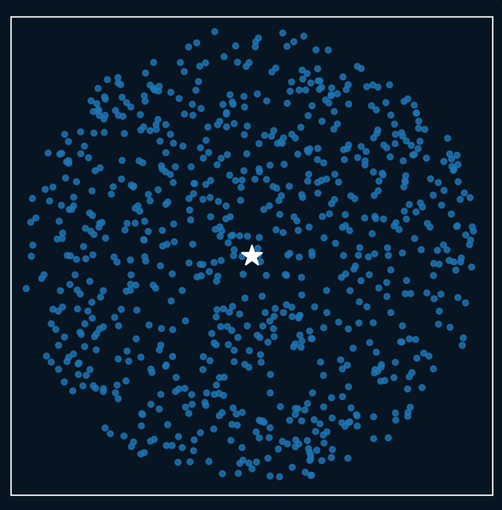
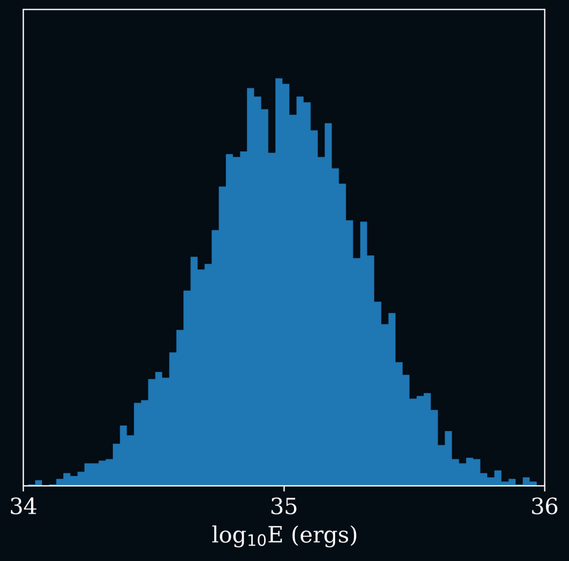
BEAMING IN FAST RADIO BURSTS

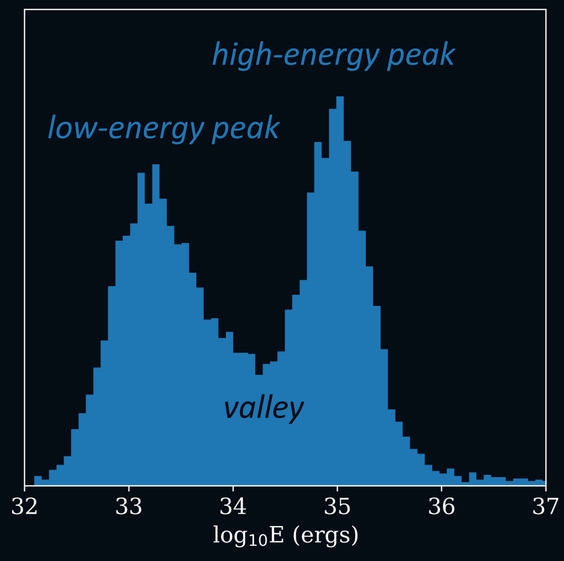
Clarinda Montilla[1] and Joscha Jahns-Schindler[2,3]
contact email: clarinda.montilla@pg.canterbury.ac.nz

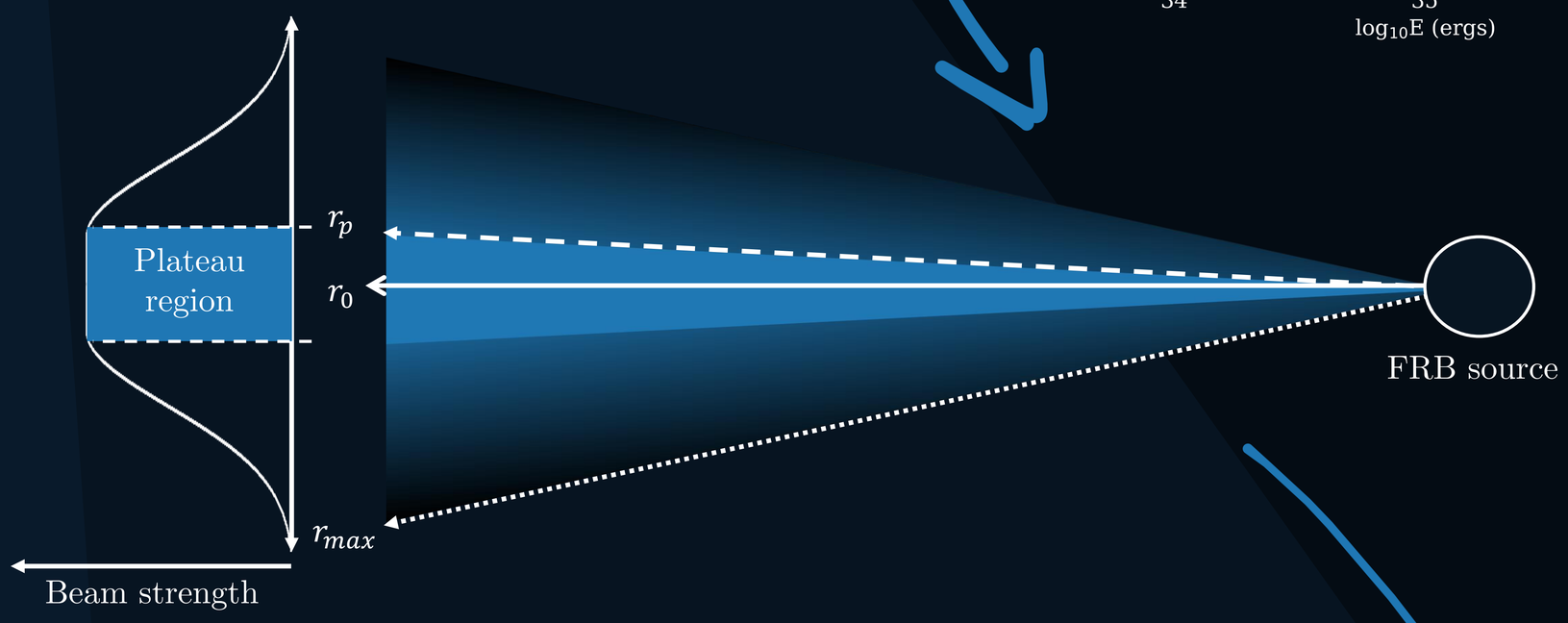
Fast Radio Bursts (FRBs) are energetic millisecond transients. Our current understanding of these events relies on the interpretation of their observed features such as energy distribution. At least three known repeater FRBs have had windows of activity where their **energy distribution is bimodal**. We recreate the energy distributions of FRB20201124A and FRB20121102 to demonstrate how beaming may have influenced observations.

Generate a set of uniformly distributed observing angles → Set a beam shape & input any additional parameters → Chose an intrinsic energy distribution → Output of observed energy distribution









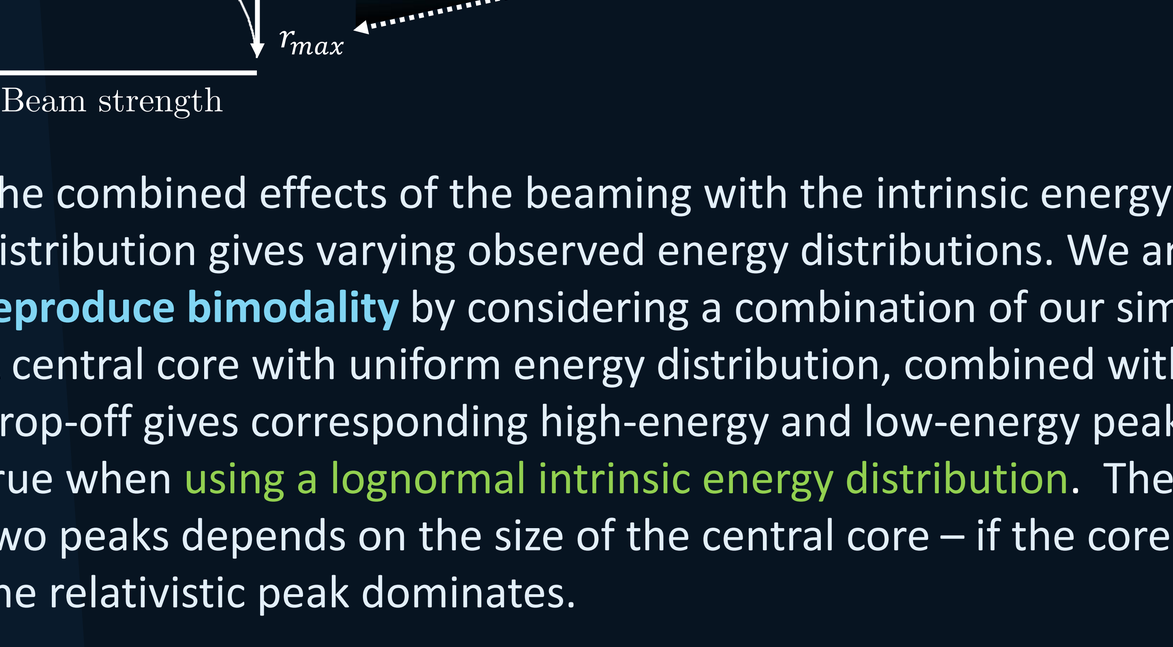
What is beaming?

Beaming is the effect of the observers angle relative to the centre of the FRB beam, on the seen energy distribution. This is *without* accounting for the intrinsic energy distribution.

The combined effects of the beaming with the intrinsic energy distribution gives varying observed energy distributions. We are able to **reproduce bimodality** by considering a combination of our simple beams. A central core with uniform energy distribution, combined with a relativistic drop-off gives corresponding high-energy and low-energy peaks. This is only true when using a lognormal intrinsic energy distribution. The height of the two peaks depends on the size of the central core – if the core is small, then the relativistic peak dominates.

What combinations did we consider?

Intrinsic energy distributions:
Single energy, power law, log normal.
***Beam shapes**:*
Isotropic, Gaussian, Gaussian & plateau, relativistic, relativistic & plateau.



We find a linear relationship between r_p and the log_{10}ratio of the height of the high-energy peak and the 'valley' between the two peaks – i.e. *the angular size of the plateau region has a power law relation to the ratio of the peak to valley*.

$$\log_{10} Ratio = (1.79 \pm 0.04) \times r_p - (0.53 \pm 0.03)$$

The energy distributions of FRB20201124A and FRB20121102 fall within our bimodal regime. This result will allow us to constrain the plateau size, the parameters of the intrinsic distribution and *possibly* the value of the Lorentz factor.

The data for FRB201124A (a) and FRB121102 (b) was available through:
(a) Zhang, Y.-K., Wang, P., Feng, Y., et al. 2022, Research in Astronomy and Astrophysics, 22, 124002, *doi: 10.1088/1674-4527/ac98f7*
(b) Li, D., Wang, P., Zhu, W., et al. 2021, Nature, 598, 267, *doi: 10.1038/s41586-021-03878-5*

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